GENERAL EPIDEMIOLOGY OF NON-COMMUNICABLE DISEASES

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BY THE END OF THE LECTURE, STUDENTS SHOULD BE ABLE TO:

- Identify chronic diseases
- Understand the magnitude of the problem
- Recognize human and financial burden
- Distinguish modifiable vs. non-modifiable risk factors
- Know preventive measures

DEFINITION OF CHRONIC DISEASES

- Chronic diseases and conditions have been variously defined.
- An EURO symposium in 1957 (1) gave the following definition .

"An impairment of bodily structure and/or function that necessitates a modification of the patient's normal life, and has persisted over an extended period of time

DEFINITION OF CHRONIC DISEASES

- **Another EURO symposium in 1965 (2) observed:**

"a chronic illness is one of long duration in which the patient is permanently incapacitated to a more or less marked degree"

- The Commission on Chronic Illness in USA has defined "chronic diseases" as "comprising all impairments or deviations from normal, which have one or more of the following characteristics
  - Permanent
  - Leave residual disability
  - Caused by non-reversible pathological alteration
  - Require special training of the patient for rehabilitation
  - Expected to require long-term supervision, observation, or care
  - Note:
    - No international consensus exists on what duration qualifies as “long-term.”
    - Many consider conditions lasting ≥ 3 months as chronic.

NON-COMMUNICABLE DISEASES

- Non-communicable diseases are the diseases that are **not transferred** from an infected person to another via any means and are mostly caused by factors like **improper lifestyle** and eating habits.

NON-COMMUNICABLE (NCDS) INCLUDE

- Cardiovascular diseases
- Renal diseases
- Nervous and mental disorders
- Musculoskeletal conditions (e.g., arthritis)
- Chronic respiratory diseases (bronchitis, emphysema, asthma)
- Permanent results of accidents
- Senility and blindness
- Cancer
- Diabetes
- Obesity
- Other metabolic and degenerative diseases
- Chronic consequences of communicable diseases

MAGNITUDE OF THE PROBLEM

Rising Importance of NCDs

- Increasing among adults in both developed & developing countries
- Leading causes of death in developed nations:
  - Cardiovascular diseases
  - Cancer

- **Global prevalence of chronic diseases is rising due to:**
  - Increased life expectancy → more vulnerable older populations
  - Lifestyle changes → higher risk behaviors
  - Improved medical care → longer survival with chronic conditions

GLOBAL IMPACT

- Over 60% of global deaths are due to NCDs
  - Highest mortality:
    - Cardiovascular diseases
    - Cancers
    - Chronic respiratory diseases
    - Digestive diseases
    - Diabetes
  - Serious consequences:
    - Loss of life & disablement
    - Family hardship & poverty
    - Economic loss to countries
- Developing nations urged to take preventive measures to avoid future "epidemics"

BURDEN OF CHRONIC DISEASES

- **Human burden:**
  - Mortality figures alone may overstate impact
  - Years of Potential Life Lost (YPLL) better for younger deaths
  - Chronic diseases often begin earlier in life
  - Quality of life measures: Quality-adjusted life years [QALY] and disability adjusted life years [DALY]) provide a clearer picture of the true impact on life quality and disability
  - Overall human cost is enormous and rising

- **Financial Burden**
  - Death costs less; living with chronic disease is expensive
  - Over 75% of U.S. Medicare spending on chronic disease care
  - New treatments improve outcomes but raise expenses
  - Indirect costs (productivity loss, absenteeism, presenteeism) often exceed direct costs

COMMON ELEMENTS IN PATHOGENESIS (RISK FACTORS):

- **Non-modifiable risk factors**
  - **Age**
  - **Sex**
  - **Genetic factors**
  - **Ethnicity**

MODIFIABLE RISK FACTORS

- **Tobacco** → 7.5 million deaths annually (including second-hand smoke)
- **Insufficient physical activity** → ~3.2 million deaths annually
- **Harmful use of alcohol** → ~3.3 million deaths annually (~4% of global deaths)
- **Unhealthy diet** →
  - **High salt** → ↑ blood pressure & cardiovascular risk
  - **Saturated & trans fats** → ↑ heart disease

- **Raised blood pressure**
  - Major risk factor for cardiovascular disease
- **Overweight & obesity**
  - ↑ risk of heart disease, stroke, diabetes
- **Higher BMI** also linked to certain cancers
  - Raised cholesterol
  - Increases risk of heart disease and stroke

- **Cancer-associated infections**
  - **Human papillomavirus (HPV)**
  - **Hepatitis B virus (HBV)**
  - **Hepatitis C virus (HCV)**
  - **Helicobacter pylori**

  Largely preventable (vaccination, safe practices) or treatable

- **Environmental risk factors**
  - **Occupational hazards**
  - **Air and water pollution**
  - **Injuries (e.g., destructive weapons, accidents)**

PREVENTION OF CHRONIC DISEASES

- 80% of chronic diseases are preventable with existing measures
- Even genetic risks can be modified through lifestyle interventions
- Prevention is the most effective long-term strategy

SOME PREVENTION AND CONTROL TASKS FOR NON-COMMUNICABLE DISEASES:

- **Tobacco avoidance** → avoid all forms, including secondhand smoke
- **Healthy diet** →
  - **Reduce** saturated fats, sweets, fried foods, red/processed meats, full-fat dairy
  - **Increase** fruits, vegetables, whole grains
  - **Limit salt** intake to ≤ 1 teaspoon/day

- **Exercise regularly** → lowers risk of colon, breast, and reproductive cancers
- **Avoid obesity** → reduces risk of multiple cancers and chronic diseases
- **Limit alcohol** → lowers risk of cancers (mouth, liver, colon, breast, etc.)
- **Avoid unnecessary radiation exposure**

- **Avoid exposure** to industrial/environmental toxins (asbestos, benzene, PCBs)
- **Avoid infections that contribute to cancer** (HBV, HCV, HIV, HPV) via vaccination & safe practices
- **Reduce stress** → chronic stress damages cardiovascular health
- **Get good quality sleep** → poor sleep linked to obesity, hypertension, diabetes, depression
- **Regular health screenings** → blood pressure, cholesterol, diabetes, cancer

Thank you